

ANIRUDDHA CHOWDHURY, Ph.D., EIT

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EDUCATION

Doctor of Philosophy in Micro and Nanoscale Systems

Mar 2021 - May 2026

Louisiana Tech University, Ruston, LA, USA

Dissertation: A Comprehensive Multiscale Computational Modeling and Investigation of Asphalt-Polymer Compatibility | Advisor: Dr. Andrew Peters | GPA: 3.85/4.00

Master of Science in Mechanical Engineering

Mar 2021 - Aug 2023

Louisiana Tech University, Ruston, LA, USA | GPA: 3.92/4.00

Bachelor of Technology in Mechanical Engineering

Aug 2012 - Jun 2016

National Institute of Technology (NIT) Jamshedpur, India | Fully funded merit scholar

SUMMARY

- **5+ years as computational lead** on a Federal Highway Administration (FHWA) funded project, developing atomistic and coarse-grained models to predict the compatibility of polymer-modified asphalt under different service conditions.
- **3 original asphalt modeling frameworks** published or under review; the compatibility framework has been applied by an independent research group in Türkiye, which led to an ongoing international collaboration, and by an industry engineer in a polymer-modified binder project.
- **Licensed Engineer Intern**, State of Louisiana (EI0036277), with the P.E. (Mechanical) exam passed.
- **Computational collaborator** on a NASA EPSCoR-funded, multi-institutional project with LSU, Tulane University, and NASA Glenn Research Center.
- **4+ years of industry experience** in chemical process engineering and automotive quality control.

Key Specialization: Multiscale molecular simulation, polymer-modified and recycled-plastic asphalt binders, asphalt oxidative aging, rejuvenators, and sustainable pavement materials.

RESEARCH PROFILES

Google Scholar: <https://scholar.google.com/citations?user=mviTyTkAAAAJ&hl=en>

GitHub: <https://github.com/anibonny93/MD-simulation-of-asphalt-and-polymer>

LinkedIn: <https://www.linkedin.com/in/aniruddha93/>

SKILLS

Molecular Simulation	LAMMPS, MARTINI 3 coarse-graining, HOOMD-blue, Polymatic, VMD, OPLS-AA force field, NEMD (SLLOD), thermodynamic integration, Flory-Huggins χ analysis
Programming and Data	Python, MATLAB, SQL, Linux/HPC job management (LONI), JMP, Minitab, Power BI
Material Characterization	FTIR, SEM with EDS, AFM, XPS, TGA, DSC, LSCM
Microfabrication	Photolithography, wet and dry etching, CVD and thin-film deposition, four-point probe I-V and C-V
Process and Design	Aspen Plus, SolidWorks, AutoCAD, Revit, DMAIC, root cause analysis, FMEA, Lean, Kaizen

ACADEMIC AND PROFESSIONAL EXPERIENCE

Graduate Research Assistant, Institute for Micromanufacturing, Louisiana Tech University

Mar 2021 - May 2026

Funding Agency: U.S. Department of Transportation, Federal Highway Administration (FHWA) (**Grant No. 693JJ320C000024**)

- Led the computational thrust of the project, using molecular simulation to identify recycled plastics that blend with asphalt binder without phase separation (see Research Projects).
- Coordinated simulation and laboratory work within the research team, and validated simulated mixing by FTIR.
- Disseminated project findings through journal publications, conference and symposium proceedings, and annual project reports.

Funding Agency: Louisiana Board of Regents (**Grant No. LEQSF(2020-23)-RD-A-18**)

- Supported simulation setup and data analysis for block copolymer grafted nanoparticle research (see Research Projects).

Funding Agency: NASA EPSCoR

- Serve as computational collaborator on a multi-institutional project with LSU, Tulane University, and NASA Glenn Research Center (see Research Projects).

Process Engineer (Production Officer), Hydrogen Peroxide Plant, Al-Razi Chemical Complex Ltd., Bangladesh Jan 2018 - Dec 2020

- **Process Modeling:** Modeled the hydrogen peroxide production process in Aspen Plus for material and energy balance analysis.
- **Process Improvement:** Raised distillation unit process capability from Cpk 1.14 to 1.40 using Six Sigma and root cause analysis.
- **Operations:** Supervised 25+ operators, maintained SOPs and safe chemical handling, monitored DCS, and reported daily KPIs to senior engineers.

Assistant Mechanical Engineer, Quality Control, AFTAB Automobiles Ltd., Bangladesh

Aug 2016 - Dec 2017

- **HVAC Validation:** Validated bus HVAC sub-assemblies through leak, pressure-integrity, and airflow testing, releasing about 30 units per month at 100% first-pass yield.
- **Polymer Manufacturing QC:** Supervised quality validation of 3,000+ injection-molded seat assemblies and fiber-reinforced plastic compression molding.
- **Design and Documentation:** Modeled chassis and prototype components in SolidWorks and AutoCAD; prepared BOMs and tender documents.

RESEARCH PROJECTS

Investigation of Asphalt-Polymer Compatibility Using Molecular Dynamics Simulation

2021 - 2024

(Supported by FHWA, U.S. DOT)

- Designed and executed all-atom MD simulations in LAMMPS to evaluate the thermodynamic miscibility of asphalt with waste and commercial polymers.
- Formulated a Flory-Huggins framework to quantify interaction parameters (χ) and predict phase behavior across blend compositions.
- Established a compatibility ranking that allows practitioners to eliminate incompatible blends before laboratory and field trials.
- Validated simulated mixing behavior experimentally by FTIR spectroscopy of laboratory-prepared blends.
- Disseminated findings in The Journal of Physical Chemistry B (2024) and two AIChE Annual Meetings.

Development of a Revised Asphalt Oxidation Model and Its Implications for Polymer Compatibility

2023 - 2026

(Supported by FHWA, U.S. DOT)

- Developed a unified oxidative aging model that links oxygen uptake to the aging state of asphalt binder.
- Introduced orbital electronegativity as a computationally efficient proxy for quantum chemical methods to predict site-specific oxidation.
- Extended the 12-component asphalt model across five aging states to represent binder chemistry over its service life.
- Quantified how oxidative aging shifts asphalt-polymer compatibility using Flory-Huggins χ parameters.
- Published in Fuel (2026); presented at the Gulf Coast Polymer Conference (2024) and TRB annual Meeting (2026).

Development of a Coarse-Grained Asphalt Model Based on the MARTINI 3 Force Field

2023 - 2026

(Supported by FHWA, U.S. DOT)

- Developed a novel coarse-grained model of virgin and oxidized asphalt, extending simulations to microsecond time scales inaccessible to all-atom models.
- Validated the model against experimental density, coefficient of thermal expansion, self-diffusion, and solubility in toluene and heptane.
- Investigated dissolution and phase separation of PE, PP, PB, PS, PET, and SBS in asphalt binder.
- Manuscript under review in the Journal of Molecular Liquids; presented at the TRB annual meeting (2024).

Molecular Dynamics Investigation of Bio-Based and Waste-Derived Asphalt Rejuvenators

2026 - Present

(International collaboration with Dr. Ahmet Münir Özdemir, Bursa Technical University, Türkiye)

- Lead the computational work of a joint experimental-computational program on waste tallow, tire pyrolysis oil, and bio-based polyurethane asphalt modifiers, with the partner laboratory supplying GC-MS, FTIR, and rheological data.
- Developed molecular models of a waste tallow rejuvenator from GC-MS composition, representing it by its dominant fatty acid methyl esters.
- Mapped SARA-rejuvenator interactions with radial distribution functions, showing the strongest rejuvenator affinity for asphaltenes, followed by aromatics.

Multi-Institutional Collaboration: Particle-Based Computational Modeling

2025 - Present

(Collaboration with Louisiana State University, Tulane University, and NASA Glenn Research Center; funded through NASA EPSCoR program)

- Develop particle-based interaction models for the project's materials systems.
- Run large-scale simulations on high-performance computing clusters.
- Analyze structural and thermodynamic simulation output and interpret it in terms of fundamental physical principles.

Domain Spacing in Lamellae Forming Block Copolymer Grafted Nanoparticles by Thermodynamic Integration

2022 - 2025

(Collaboration with J. Wiley; supported by the Louisiana Board of Regents, award LEQSF(2020-23)-RD-A-18)

- Supported simulation setup and data analysis for a thermodynamic integration study of lamellar domain spacing, run on LONI high-performance computing resources.
- Co-authored the resulting manuscript, forthcoming in the Journal of Molecular Liquids.

PEER REVIEWED PUBLICATIONS

1. **Chowdhury, A.**, Nourian, P., Wasiuddin, N., & Peters, A., 2026, "Development of a revised molecular model for asphalt oxidation and its implications for polymer compatibility," *Fuel*, 417, 138564.
2. **Chowdhury, A.**, Nourian, P., Wasiuddin, N., & Peters, A., 2024, "Investigation of polymer-asphalt compatibility using molecular dynamics simulation," *The Journal of Physical Chemistry B*, 128(19), 4821-4829. (Top 10% most cited, 2024)

Manuscripts Under Review / In Preparation

1. **Chowdhury, A.**, Nourian, P., Wiley, J., & Peters, A., "Martini based coarse-grained modeling and analysis of polymer-asphalt dissolution dynamics," *Journal of Molecular Liquids* (under review).
2. Wiley, J., **Chowdhury, A.**, Nourian, P., & Peters, A., "Analysis of domain spacing in lamellae forming block copolymer grafted nanoparticles by thermodynamic integration," *Journal of Molecular Liquids* (forthcoming).
3. **Chowdhury, A.**, et al., "Evolution of asphalt molecular models: a review", in preparation for *Fuel*.

CONFERENCE PRESENTATIONS

1. **Chowdhury, A.**, Peters, A., Nourian, P., & Wasiuddin, N., 2026, "Investigating the effect of asphalt-polyethylene compatibility using molecular dynamics simulation," *Transportation Research Board (TRB) 105th Annual Meeting*, Washington, D.C., Jan 2026.
2. **Chowdhury, A.**, Nourian, P., & Wasiuddin, N., 2024, "Investigation of phase separation behavior of polymer-asphalt mix using molecular dynamics simulation," *Gulf Coast Polymer Conference*, University of Southern Mississippi, Hattiesburg, MS, Dec 2024.
3. Peters, A., & **Chowdhury, A.**, 2024, "A coarse-grained model for asphalt binder and polymer mixtures based on the MARTINI force field," *TRB 103rd Annual Meeting*, Washington, D.C., Jan 2024.
4. **Chowdhury, A.**, Peters, A., Nourian, P., & Wasiuddin, N., 2023, "Investigation of waste plastic-asphalt compatibility using MD simulation," *AIChE Annual Meeting*, Orlando, FL, Nov 2023.
5. Peters, A., **Chowdhury, A.**, Rahman, R., Shafe, A., Hossain, R., & Wasiuddin, N., 2021, "Use of waste plastic in asphalt binder: theoretically and experimentally justified identification of compatible blends," *AIChE Annual Meeting*, Boston, MA, Nov 2021.

PAPER REVIEW

- Transportation Research Board (TRB) Annual Meeting
- RSC Advances (Royal Society of Chemistry)
- The Journal of Physical Chemistry (ACS Publications)
- ACS Omega (ACS Publications)
- International Journal of Pavement Research and Technology (Springer)

Total manuscripts reviewed: 14

AWARDS AND HONORS

- Outstanding Graduate Student Award, College of Engineering and Science, Louisiana Tech University (2025)
- 1st Place, Oral Presentation, Graduate Research Symposium, Louisiana Tech University (2023 and 2024)
- 3rd Place, Oral Presentation, Graduate Research Symposium, Louisiana Tech University (2025)
- Harry J. Gaston, Jr. Memorial College of Engineering and Science Scholarship (2025)
- Adamantios K. Gorgolis Graduate College of Engineering and Sciences Scholarship (2025)
- Kaizen Award, Al-Razi Chemical Complex Ltd. (process capability improvement)
- Fully funded merit scholarship for undergraduate study, NIT Jamshedpur

US NATIONAL INTEREST

My research develops computer models that predict whether recycled plastics and bio-based additives can be safely blended into asphalt before any road is paved.

- **Federal priority:** My doctoral research was funded by the Federal Highway Administration to address durability and sustainability challenges on the nation's highways.
- **Longer-lasting roads:** Oxidative aging is a leading cause of asphalt cracking. My published aging model predicts how binder chemistry and polymer compatibility change over a pavement's life, supporting more durable designs.

- **Waste plastic diversion:** Screening compatible plastics by simulation helps turn landfill-bound plastic into a pavement modifier without the phase separation that causes early failure.
- **Lower development cost:** Simulation screening replaces much of the trial-and-error blending that binder producers and DOTs now rely on.
- **Safer emissions profile:** Concerns remain about fuming and the release of volatile organic compounds (VOCs) and polyaromatic hydrocarbons from plastic-modified roads. Identifying stable, compatible blends by simulation supports the laboratory benchmarking and standards needed before agencies adopt plastic roads.

LEADERSHIP AND MENTORING

- Ex-Officio Senator, Student Government Association, Louisiana Tech University
- President, Students' Association of Bangladesh
- Mentored graduate and undergraduate students in molecular dynamics, polymer chemistry, and data analysis.

LICENSURE, CERTIFICATIONS, AND AFFILIATIONS

- Licensed Engineer Intern, State of Louisiana (EI0036277), P.E. (Mechanical) exam passed
- Six Sigma Green Belt | Lean Manufacturing Certification | PMP Training
- Member, American Institute of Chemical Engineers (AIChE)
- Member, Project Management Institute (PMI)

REFERENCES

Dr. Andrew Peters

Terrell and Henry Families Associate Professor
Chemical Engineering, Louisiana Tech University
Email: apeters@latech.edu

Dr. Jack Youtcheff

Principal Infrastructure Materials Research Engineer
Federal Highway Administration
Email: jack.youtcheff@dot.gov

Dr. Nazimuddin Wasiuddin

Professor and MCI Endowed Professor
Mississippi State University
Email: wasi@cee.msstate.edu

Dr. Md Shams Arafat

Transportation Asset Management Engineer
California Department of Transportation (Caltrans)
Email: md.shams.arafat@dot.ca.gov